

TurtleStitch in the classroom
What to do if USB ports are blocked?
Presentation for discussion

Agenda

- 1) The problem: Inventory of limitations around .DST file exchange (via USB sticks or otherwise)
- 2) Inventory of potential solutions and limitations
- 3) What could be useful solutions to further explore in detail. Which options can work for you?
- 4) Next steps

Part 1

The problem

The problem

Increasingly, teachers using TurtleStitch in schools are confronted with new limitations

- 1) USB ports can no longer be used for USB sticks as storage device (AMP)
- 2) Files can only be put in the Downloads folder (H)
- 3) No executables can be run, not even from the Downloads folder (AP)
- 4) Only one prescribed Wifi access point can be accessed (PmA)
- 5) No Bluetooth can be used for file sharing (P)
- 6) Embroidery machines with Wifi access are not allowed in the network (PM)
- 7) In e.g. China, access to internet is blocked in some cases (S)
- 8) Externals / teachers cannot access the school's Wifi (MP)
- 9) Cannot send mail with .DST attachments (even when zipped and encrypted)

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Lower case =
some restrictions
apply

Questions:

- a) Are there more obstacles observed?
- b) Who encounters which obstacle?
- c) Does downloading to micro:bit via the browser still work for all? It uses WebUSB
- d) Same question for Arduino?

Part 2

Options for a solution

Ways around

- This presentation contains some ideas how to go around the restrictions
- Some ideas are even wilder than others
- And some / many ideas may not be possible / practical at all
- It is the intent to spark a brainstorm
- After that we can decide which tracks to further explore / work out

Options in this presentation

- Option 1: convince IT to accept USB sticks in a limited way.
- Option 2: use DropStitch: transfer via Bluetooth to a Rpi Pico which acts as a USB stick
- Option 3: Use Bitlocker encrypted USB sticks with a PC in between to copy to a plain USB stick
 - Option 3a: Use a Raspberry Pi instead of a PC
- Option 4: Copy .XML via the TurtleStitch cloud
 - Option 4a: Use a local NAS server instead of the cloud
- Option 5: Copy .DST files via a cloud
 - Option 5a: Use a local NAS server instead of the cloud
- Option 6: Create a device that talks WebUSB to the PC and accepts a plain USB stick on the other side
- Option 7: Create a device that is Bitlocker stick when in the PC and a plain stick when in the machine
- Option 8: Encode the .DST file into an audio file with FSK, play over the audio interface and create a device that decodes this into a file on a USB stick
- Option 9: Send the (potentially encrypted zipped) .DST file via email to the teacher

Option 1: Convince your IT to open the USB ports selectively

- Whether this is possible for a subset of PCs and for specific USB devices depends on the security software that is used
- Although there are indications that it is possible with some Microsoft solutions
- It can be further investigated how this works if your IT does not know.

Can be done now

Option 2: Use DropStitch

- Have RPi ZeroW connected to the embroidery machine as if it were a USB stick (2 computers connect to RPi, 1 embroidery machine)
- Send .DST files over Bluetooth to the RPi
- Send a special file to terminate the transfer or to empty the “virtual USB stick” (the RPi).
- Created by Magaret’s brother
- Can be found via <https://dropstitch.design.blog/>
- Only solution for cases where there is filesharing and teachers laptop has USB blocked (as otherwise .DST files can be transferred via the network)
- For Windows. Also for Mac and Linux

Can be used now

Limitations

- Bluetooth must be existent and be permitted to be used for filetransfer on the PCs
- Number of PCs (2) / number of machines addressable (1). So not usable from every classroom laptop, but could be used from the teacher's PC if they get the designs via a file sharing or cloud.
- Startup time? Reset time 20s
- RPI zero W difficult to get at times, currently OK

Considerations for the other options

- The only thing we know for sure is that an .html page with JavaScript that can run in the browser will be working in any environment
 - If it is not then TurtleStitch as such cannot run either
- Make use of interfaces that are unlikely to be blocked because they are used for core functionality of the schools
- We want to minimize the amount of work for the teacher
- We want to minimize the amount of steps for the participant
- We want to not have the cost for extra devices or software explode

Option 3 – Use Bitlocker USB sticks

- Get one or more Bitlocker encrypted sticks from your IT department, including the key to unlock
 - Assuming that BitLocker USB sticks are allowed
- When encrypted USB sticks are allowed then*:
 - Put the .DST file on that encrypted USB stick
 - Open the encrypted USB stick on a private laptop
 - A key needs to be typed in
 - Copy the .DST file to another non-encrypted USB stick
 - Put that unencrypted USB stick in the machine

* Not tested, but I believe it works like this

Can it be used now?

Option 3 - details

- Unclear whether alternatively own bitlocker encrypted USB sticks can be used or whether IT needs to supply them. But it looks that the user can encrypt an empty stick with an own password.
- Bitlocker is not supported on Windows Home, but there is a package from [Hasleo](#). Even available for Windows 7 and up. Also available for Mac and Linux
 - Unclear whether it could be used on Raspberry Pi
- We can potentially automate the process on the teacher PC or dedicated PC. Like:
 - Put encrypted USB stick in
 - Put regular USB stick in
 - Automatically move the .DST file
 - Eject both USB sticks
 - Give a message that the sticks can be removed

Option 3a

- Potentially use Raspberry Pi for the process of option 3
- Advantages:
 - more affordable, although an old laptop (Windows 7 and up) may be cheaper
 - Less volume to carry

Issues with option 3

- Not clear whether encrypted USB disks are always allowed and available
 - Margaret is not allowed to use encrypted USB sticks
- Hassle, work for the teacher

Option 4 – via the TurtleStitch cloud

- Use a TurtleStitch group account
- Let the participant upload the .XML file to TurtleStitch Cloud
- Let the teacher on a private laptop download the .XML file
- Open the .XML file, inspect it, let it run and generate the .DST file
- Put the .DST file on a regular USB stick
 - Potentially use TSUSB-Export to make it easier and do checks
- Put the USB stick in the embroidery machine
- There may be opportunities to partly automate the actions.

Can be used now

Issues with option 4

- If the design is composed of a number of independently running TurtleStitch stacks (which should not be the case, but ...) then the design cannot be embroidered as the .DST file cannot be recreated on the teacher's laptop
- It takes time to open, run the design, which in case of filled fonts may take quite a while, and then export to .DST and copy the .DST file to a USB stick
 - All that time the teacher's laptop is unavailable for other work, e.g. the design of another participant

Option 4a: Use file share i.s.o. TS Cloud

- Use file sharing that may exist in the school
- Copy .XML via a file sharing

Option 5: .DST file via TurtleStitch cloud or other server

- Introduce an option to store the .DST file in the TurtleStitch cloud or another server in the internet or a local network share. Do so in addition to the .XML file
 - Could we do it via a custom block in TS?
- Download the .DST file on a teachers PC and there write it to a USB stick.
- USB stick to the embroidery machine
- See the detailed proposed workflow at the end of the presentation.

Option 5a: Use network share i.s.o. TS Cloud

- Use a network share
- Copy .DST via a network share

Option 6: WebUSB

- It is a method by which browsers can access USB devices.
- It is the way micro:bit (and probably Arduino) can be loaded from a browser
- See <https://en.wikipedia.org/wiki/WebUSB>
- We can potentially create a USB device based on e.g. Arduino or Raspberry Pi (Pico) that speaks WebUSB at one end and can accept a USB stick at the other

WebUSB useful links

- <https://github.com/webusb/arduino>
- <https://crates.io/crates/usb-device>
- <https://dfu-util.sourceforge.net/>
- <https://developer.chrome.com/docs/capabilities/usb>
- <https://windowsforum.com/threads/unlocking-web-to-device-connectivity-in-microsoft-edge-with-web-bluetooth-and-webusb.375384/>
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Option 7: Create device that is Bitlocker at one end and a regular USB stick at the other

- Implement Bitlocker on Arduino or RPi
- When inserted in the PC it presents itself as Bitlocker
- When inserted in the embroidery machine it presents itself as a normal FAT16 drive
- The .DST file is written on the PC and then read on the machine

Option 7 questions and issues

- How is the Bitlocker key known?
 - It looks that this can be specified when a blank USB stick is inserted
- Is the Bitlocker key diversified (different per PC)?
 - No, since you can choose your own key
- Not usable for cases where the teacher has signed a contract not to use file export to USB sticks even via encrypted sticks.

Option 8: Encode .DST file as audio file. Play to a device that puts it on USB Stick

- Create a JavaScript that encodes the .DST file into an audio file with FSK (Frequency Shift Keying).
- Play the file on the audio interface
- Create a device that decodes the audio file into a .DST file.
- Copy the .DST file onto a USB stick
- Put the USB stick into the machine.

Needs development

Option 9: email the .DST file to the teacher

- Let the participant zip the .DST file (and potentially encrypt if needed to not be trapped)
- This could be automated via a .HTML page with JS
- Teacher downloads the file to a USB stick (Use personal laptop if no USB access permitted)

Limitations and issues

- When not allowed to send any attachment then the .DST file could be encoded into the body text.

Part 3

What could be useful
solutions to further explore
in detail.

Which options can work for you?

What options would work for you?

- Option 1: convince IT to accept USB sticks in a limited way.
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Part 4

Next steps

Conclusion

- Option 5 (.DST via TS Cloud or network share) looks the most promising for now

Next steps

- Simon and Hans will explore option 5

BACKUP

Preferred use case for Option 5 – initial design

- Participant makes the design
- Participant hits a single button in TurtleStitch that will
 - Save the design in a .XML file
 - Prompt for a name if no save was done before
 - Export the .DST file
 - Check the design for certain configurable characteristics (see further on in the presentation)
 - Reports errors if characteristics are not OK. Teacher will have a password protected option to override the errors.
 - When OK or accepted TS sends the .XML and .DST file to a location on the internet or a network share and tells the participant to fetch the USB stick with the files from the teacher.
- Teacher has a number of (one or more) USB sticks in a teacher laptop (with USB access), ready to be used
 - Teacher uses own laptop connected via e.g. mobile phone to internet if connection to school Wifi is not possible / USB ports on teacher laptop are also blocked.
- The .DST and .XML file are automatically downloaded from the location on the internet to one of the free USB sticks and then that USB stick is automatically ejected.
- When download is complete, the teacher will get a pop up to tell which USB stick to take out, plus the name of the file (which should better be the name of the participant)
- The participant will get the USB stick to take to the embroidery machine
 - In case of a USB connected embroidery machine, the teacher will send the .DST file to the machine when the machine is free.

Preferred use case - rework

- After the work has been embroidered, the participant
 - Takes the embroidery home and puts the use USB stick into a tray of used sticks for cleaning, or
 - Goes back to a PC, often – but not always - the original PC and
 - Gets the .XML file from the internet location or network share and drops it into an open TS window
 - Updates the design and the process repeats

Cleanup

- At the end or when running out of empty USB sticks
 - The teacher inserts the stick into either 1) the PC used for exporting to USB sticks (as used before) or into 2) a dedicated PC
 - In both cases the USB stick will be automatically cleaned: the TS related content will be erased (see the process of TSUSB-Cleanup). Potentially a log item can be created with the .XML and .DST file for later processing (e.g. send to a participant or study a weird case).

Error cases

- If the copying to the USB stick goes wrong, the teacher must have the option to restart the copy
- If the copy by the participant to the internet or file share goes wrong, the participant repeats the process

Architecture for an option 3

- Have a web page with a JS program that
 - Can be clicked after the exporting of the .DST file (better in the same process as TS)
 - Does basic checks
 - Sends the .DST file to the TS cloud, other server in the internet or local file share
- Alternatively, would it be possible to create TS custom block that does this?
- Have a program running in the background on the teacher's private laptop* that copies or moves the .DST file to a USB stick that is connected to that laptop
 - Signals that the USB stick can be taken out once the copying is ready

* Whoever owns the laptop is not the matter, as long as it has no USB restrictions and can connect to the internet – via the school's wifi or via a mobile connection

Requirements for the JS program

- Check whether there is a .DST file present
- Check the .DST file for:
 - Size
 - Amount of stitches
 - Amount of colors
 - Whether there is a likely corresponding .XML file
 - Whether there .XML and .DST files are recently generated
 - When possible whether there are density warnings (only possible when further integrated with TurtleStitch)
 - Take the values from a .ini file in the Downloads folder or from a location in the group account

* These are the current checks in TSUSB-Export